

Zachary McGraw

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Research Focus

I am an atmospheric physicist working to advance understanding of how aerosol and cloud processes shape Earth's climate system. My research uses physical theory and a hierarchy of models to close gaps in the mechanistic understanding of interactions among microphysical processes, radiation, convection, atmospheric dynamics, precipitation, and temperature.

Education

Ph.D. in Geosciences

12/2020

University of Oslo | Department of Geosciences

Supervisor: Trude Storelvmo

Thesis title: *Global radiative impacts of aerosol variations through mixed-phase and cirrus clouds*

M.Phil. in Atmospheres, Oceans & Climate Dynamics

05/2017

Yale University | Department of Geology & Geophysics

Supervisor: Trude Storelvmo

B.Sc. double-majoring in Physics, Applied Math & Statistics

05/2012

Johns Hopkins University

Work Experience

Postdoctoral researcher at the University of Vienna

02/2021 – present

Supervisors: Blaž Gasparini & Aiko Voigt

- Research on ice cloud seeding and climate
- Fundamental research on tropical convection and clouds

Postdoctoral research scientist at Columbia University & NASA GISS

01/2021 – 01/2026

Supervisor: Lorenzo Polvani

- Research projects on how stratospheric aerosols influence temperature and precipitation
- Continued research on the role of clouds in Earth's climate system.
- Development tasks improving aerosol treatment in NASA GISS ModelE general circulation model

Research assistant at Columbia University & NASA GISS

10/2012 – 07/2014

Supervisors: Susanne Bauer & Kostas Tsigaridis

– Programming project to automate reading of aerosol measurements for model input and visualization

– Climate model validation tasks using flight campaign measurements

Visiting undergraduate researcher at the University of Galway

06/2010 – 07/2010

within the Centre for Climate and Air Pollution Studies (C-CAPS) under Prof. Colin O'Dowd

– Aided in modeling the Eyjafjallajökull volcanic ash plume that disrupted European flights in 2010

Peer-Reviewed Publications

McGraw, Z., & Polvani, L. M. (2026). Direct Radiative Impacts of Stratospheric Aerosols on the Tropical Troposphere: Clouds, Precipitation, and Circulation in Convection-Resolving and Global Simulations. *Journal of Climate*. Published online 7 July 2026.

<https://doi.org/10.1175/JCLI-D-25-0688.1>

McGraw, Z., & Polvani, L. M. (2025). Do Climate Models Support Claims of Volcanic Global Catastrophes?. *Geophysical Research Letters*, 52(18), e2025GL117611.

<https://doi.org/10.1029/2025GL117611>

Raiter, D., **McGraw, Z.**, & Polvani, L. M. (2025). Non-linear and distinct responses of temperature and precipitation to volcanic eruptions with stratospheric sulfur injection from 5 to 160 Tg.

Geophysical Research Letters, 52(20), e2025GL116372. <https://doi.org/10.1029/2025GL116372>

McGraw, Z., Polvani, L. M., Gasparini, B., Van de Koot, E. K., Voigt, A. The Cloud Radiative Response to Surface Warming Weakens Hydrological Sensitivity. *Geophysical Research Letters*, 52, e2024GL112368. <https://doi.org/10.1029/2024GL112368>

Gjelsvik, A. B., David, R. O., Carlsen, T., Hellmuth, F., Hofer, S., **McGraw, Z.**, Sodemann, H., & Storelvmo, T. (2025). Using a region-specific ice-nucleating particle parameterization improves the representation of Arctic clouds in a global climate model. *Atmospheric Chemistry and Physics*, 25(3), 1617-1637. <https://doi.org/10.5194/acp-25-1617-2025>

Hofer, S., Hahn, L. C., Shaw, J. K., **McGraw, Z.**, Bruno, O., Hellmuth, F., Pietschnig, M., Mostue, I.A., David, R.O., Carlsen, T. & Storelvmo, T. (2024). Realistic representation of mixed-phase clouds increases projected climate warming. *Communications Earth & Environment*, 5(1), 390.

<https://doi.org/10.1038/s43247-024-01524-2>

McGraw, Z., & Polvani, L. M. (2024). How volcanic aerosols globally inhibit precipitation.

Geophysical Research Letters, 51(13), e2023GL107930. <https://doi.org/10.1029/2023GL107930>

McGraw, Z., DallaSanta, K., Polvani, L. M., Tsigaridis, K., Orbe, C., & Bauer, S. E. (2024). Severe global cooling after volcanic super-eruptions? The answer hinges on unknown aerosol size.

Journal of Climate, 37(4), 1449-1464. <https://doi.org/10.1175/JCLI-D-23-0116.1>

McGraw, Z., Storelvmo, T., Polvani, L. M., Hofer, S., Shaw, J. K., & Gettelman, A. (2023). On the links between ice nucleation, cloud phase, and climate sensitivity in CESM2. *Geophysical Research Letters*, 50(17), e2023GL105053. <https://doi.org/10.1029/2023GL105053>

- Gettelman, A., Morrison, H., Eidhammer, T., Thayer-Calder, K., Sun, J., Forbes, R., **McGraw, Z.**, Zhu, J., Storelvmo, T., & Dennis, J. (2023). Importance of ice nucleation and precipitation on climate with the Parameterization of Unified Microphysics Across Scales version 1 (PUMASv1). *Geoscientific Model Development*, 16(6), 1735-1754. <https://doi.org/10.5194/gmd-16-1735-2023>
- Zhu, J., Otto-Bliesner, B. L., Brady, E. C., Gettelman, A., Bacmeister, J. T., Neale, R. B., Poulsen, C. J., Shaw, J. K., **McGraw, Z.**, & Kay, J. E. (2022). LGM paleoclimate constraints inform cloud parameterizations and equilibrium climate sensitivity in CESM2. *Journal of Advances in Modeling Earth Systems*, 14(4), e2021MS002776. <https://doi.org/10.1029/2021MS002776>
- Shaw, J., **McGraw, Z.**, Bruno, O., Storelvmo, T., & Hofer, S. (2022). Using satellite observations to evaluate model microphysical representation of arctic mixed-phase clouds. *Geophysical Research Letters*, 49(3), e2021GL096191. <https://doi.org/10.1029/2021GL096191>
- McGraw, Z.**, Storelvmo, T., Samset, B. H., & Stjern, C. W. (2020). Global radiative impacts of black carbon acting as ice nucleating particles. *Geophysical Research Letters*, 47(20), e2020GL089056. <https://doi.org/10.1029/2020GL089056>
- McGraw, Z.**, Storelvmo, T., David, R. O., & Sagoo, N. (2020). Global radiative impacts of mineral dust perturbations through stratiform clouds. *Journal of Geophysical Research: Atmospheres*, 125(23), e2019JD031807. <https://doi.org/10.1029/2019JD031807>
- Gasparini, B., **McGraw, Z.**, Storelvmo, T., & Lohmann, U. (2020). To what extent can cirrus cloud seeding counteract global warming?. *Environmental Research Letters*, 15(5), 054002. <https://doi.org/10.1088/1748-9326/ab71a3>
- O'Dowd, C., Varghese, C., Martin, D., Flanagan, R., McKinstry, A., Ceburnis, D., Ovadnevaite, J., Martucci, G., Bialek, J., Monahan, C., Berresheim, H., Vaishya, A., Grigas, T., **McGraw, Z.**, Jennings, S.G., Langmann, B., Semmler, T., McGrath, R. "The Eyjafjallajökull ash plume–Part 2: Simulating ash cloud dispersion with REMOTE." *Atmospheric Environment* 48 (2012): 143-151. <https://doi.org/10.1016/j.atmosenv.2011.10.037>

Manuscripts Under Review

- McGraw, Z.**, & Gasparini, B. (2026). No Reduction of Tropical Convection with Warming Expected from Theory or Models. Manuscript under review at *Geophysical Research Letters*. Preprint at <https://arxiv.org/abs/2608.14483>
- McGraw, Z.**, Gasparini, B., Mol, W., & Voigt, A. (2026). Reconciling the Lack of a Robust Anvil Cloud Amount Response to Warming. Manuscript under review at *Geophysical Research Letters*. Preprint at <https://arxiv.org/abs/2609.00429>
- McGraw, Z.**, Goldman, B. O., & Polvani, L. M. (2026). No Evidence for Strongly Enhanced Northern Hemisphere Cooling from Extratropical Versus Tropical Eruptions. *Geophysical Research Letters*, in revision.
- Mahowald, N., Liddy, H., Marlon, J., Tsigaridis, K., Winckler, G., Hamilton, D., Kok, J., Sigl, M., Abell, J., Aboagye-Okyere, A., Amooli, J. A., Albani, S., Bauer, S., Beaudon, E., Brown, H., Chellman, N., Cosentino, N., Goodkin, N., Guiterman, C., Hantson, S., He, C., Intveld, A., Kaplan, J., Klavans, J., Lambert, F., Liu, H., Loutre, M.-F., McConnell, J., **McGraw, Z.**, Mezuman, K., Miller, R., O'Mara, N., Ong, M. R., Schmidt, G., Vachula, R., Wu, L., & Zhang, B.

(2026). Characterizing Natural Aerosol Variability and its Uncertainty in the Earth System. Manuscript under review at *Atmospheric Chemistry and Physics*. Preprint at <https://doi.org/10.5194/egusphere-2026-4532>

Invited Presentations

Incorporating Clouds into Global Precipitation Change: Toward Explained and Constrained Responses to Warming and Aerosols, Joint Oslo Seminar in atmospheric, ocean and climate science, University of Oslo, Norway, 12 May 2026.

Are Volcanic Impacts on Global Climate Overblown?, Invited talk for the BOKU-Met Seminar, University of Natural Resources and Life Sciences (BOKU), Austria, 24 March 2026.

Global Impacts of Very Large Volcanic Eruptions: Lessons and Limits of Climate Modeling. Invited talk for the GMAO seminar at Goddard Space Flight Center, May 27, 2025.

The Cloud Conundrum in Climate Prediction. Invited talk for the ECSD Seminar at Brookhaven National Laboratory, February 27, 2025.

Do Volcanic Aerosols Severely Alter Global Climate? Invited talk at the American Meteorological Society annual meeting, January 16, 2025.

Are Volcanic Impacts on Global Climate Overblown? Invited talk for the Atmosphere Ocean Science colloquium at NYU Courant Institute of Mathematical Sciences, April 3, 2024.

Other Presentations

McGraw, Z., & Polvani, L. M. (2026, May). Inadvertent Tropical Rainfall Responses to Stratospheric Aerosol Injection: Disentangling Aerosol-Radiation-Precipitation Coupling in Convection-Resolving and Global Simulations. *EGU General Assembly Conference 2026*. (talk)

McGraw, Z., & Polvani, L. M. (2026, June). The Direct Radiative Influence of Stratospheric Aerosols on Precipitation: Insights from Convection-Resolving and Global Simulations. *Gordon Research Conference on Climate Engineering 2026*. (poster)

McGraw, Z. Advancing Precipitation Change Science: From Crude Models Towards Process-Aware Prediction, *Department of Meteorology and Geophysics Colloquium*, University of Vienna, Austria, 10 March 2026. (talk)

McGraw, Z., Polvani, L. M., Gasparini, B., Van de Koot, E. K., Voigt, A. Cloud Feedbacks Drive Uncertainty in Hydrological Sensitivity. *EGU General Assembly Conference 2025*. (poster)

McGraw, Z., DallaSanta, K., Polvani, L. M., Tsigaridis, K., Orbe, C., & Bauer, S. (2024, January). Super-Volcanic Eruptions May Not Cause Catastrophic Cooling. *AMS Annual Meeting 2024*. (poster)

McGraw, Z., & Polvani, L. (2023, May). Reassessing post-eruption precipitation anomalies in the context of natural variability. In *EGU General Assembly Conference 2023*. (talk)

- McGraw, Z., Storelvmo, T., & Polvani, L. M. (2022, December). The Link Between Ice Nucleation and Climate Sensitivity. *LAGU Fall Meeting 2022*. (poster)
- McGraw, Z., Dallasanta, K., Polvani, L., Tsigaridis, K., Orbe, C., & Bauer, S. (2022, May). Aerosol properties crucial to reconciling supervolcanic cooling estimates. *EGU General Assembly Conference 2022*. (talk)
- McGraw, Z., Storelvmo, T., & Smith, R. B. (2017, December). Improving the simulation of mineral dust emission in a global climate model by incorporating sub-grid scale wind phenomena. *AGU Fall Meeting 2017*. (poster)
- McGraw, Z., Smith, R., & Storelvmo, T. (2016, April). Identification and Attribution of Global Wind Speed Trends at 100m. *EGU General Assembly Conference 2016*. (poster)
- McGraw, Z. (2015, December). Trends in 100m Wind Speed using Global High-Resolution Downscaled Analysis. *AGU Fall Meeting 2015*. (poster)
- McGraw, Z., & Storelvmo, T. (2014, December). The Link Between Mineral Dust and Wind Speed: Implications for Wind Energy in the Maghreb. *AGU Fall Meeting 2014*. (poster)

Teaching Experience

Teaching fellow at Yale University

Courses:

- Atmosphere, Ocean, & Environmental Change + lab (Fall 2014)
- Physics of Weather and Climate (Fall 2015)
- Climate Dynamics (Spring 2017)
- Atmosphere, Ocean, & Environmental Change + lab (Fall 2017)

Skills

- *Climate and atmospheric model development.* Extensive experience modifying and improving atmospheric models, including aerosol and cloud microphysics in NASA GISS ModelE and CESM2. Identified fundamental model deficiencies and developed and implemented improvements in collaboration with model development teams.
- *Process-based numerical experimentation.* Experienced in designing custom model experiments to isolate physical mechanisms, including modifying coded process representations, emissions, parameterizations, and model configurations.
- *High-resolution and hierarchy-of-models simulations.* Experience with global climate models, convection-resolving models, and large-eddy simulations, including GISS ModelE, CESM, SAM, WRF, and ICON. Skilled in designing idealized experiments that isolate atmospheric processes and connect them to climate-scale responses.
- *Climate model diagnosis and analysis.* Extensive experience diagnosing model behavior using radiative kernels, process-denial experiments, energy-budget analysis, and mechanistic decomposition of climate responses, including CMIP and RCEMIP model output.

- *Physical and analytical modeling.* Analytical skills in atmospheric thermodynamics, radiation, convection, clouds, precipitation, aerosol–cloud interactions, and atmospheric dynamics, with particular emphasis on connecting mathematical formulations to their physical interpretation.
- *Scientific model evaluation.* Experienced in identifying discrepancies between models, observations, and theoretical expectations and tracing them to specific physical assumptions, parameterizations, or numerical representations.

Awards

- Co-I and wrote a large portion of a selected \$1.04 million NASA grant proposal entitled *Advancing Convective Microphysics in GISS ModelE to Improve Processes Driving Intense Convective Precipitation and Anvil Cloud Feedback*. This was among the 23% of Modeling, Analysis, and Prediction (MAP) program submissions deemed suitable for award on 2 May 2025. However, no funds were dispensed due to the current US federal government situation.
- Best NASA GISS papers 2024 | Awarded place #2 for writing the paper *Severe global cooling after volcanic super-eruptions? The answer hinges on unknown aerosol size*.

Leadership

- Led/facilitated multiple meetings at NASA GISS. This includes organizing a monthly cloud group meeting (Aug. 2024 – Jan. 2026), and co-leading with Dr. Kate Marvel a monthly discussion on geoengineering (Apr. 2024 – Jan. 2026).
- Co-mentored a total of 5 students and early-career researchers at the request of their supervisors between the University of Oslo, Columbia University, and the University of Vienna (ongoing).